

Orders of Magnitude

An immersive educational virtual reality application about the Universe

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ABSTRACT

Orders of Magnitude is an immersive educational virtual reality application. Each player can explore all the known scales in our universe by zooming in or out. This scientifically accurate experience contains elements such as galaxies, stars, the solar system, Earth, the human brain, neurons, DNA, atoms and subatomic particles. The majority of the content is taken from various scientific databases and is represented, wherever possible, as it exists in nature. An artistic impression is used to depict elements for which little or no information is available about their visual presence.

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1 INTRODUCTION

“Have you ever wondered what the entire Universe looks like? With the myriad Galaxies, including our Milky Way galaxy, containing billions of Stars, and our own Solar System? See the Earth from above, including the International Space Station and an astronaut in orbit. What does his brain look like? What about its Neural Network, all the neuron cells and their DNA? Have you seen an electron cloud surrounding the tiny atomic nucleus, filled with protons and neutrons? Have you seen what lies at even smaller scales?”

You can see an artistic visualization of all of these in the project.

2 SCIENTIFICALLY ACCURATE

Most of the content is taken from various scientific databases and is striving to represent elements as they exist in their nature. Finding the correct databases was not an easy task and a significant amount of time was needed to gather data and correctly display them.

Example of some of the databases used in this project:

- Position of nearby 40 000 Galaxies – 2MASS Redshift Survey
- Position of nearby 100 000 Stars – Hipparcos catalogue
- Brain connections – Human Connectome project

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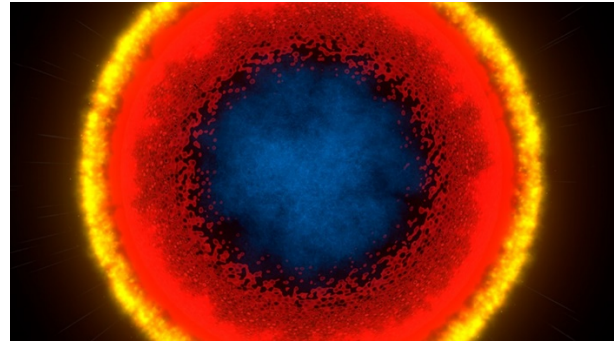


Figure 1: Observable Universe rendering

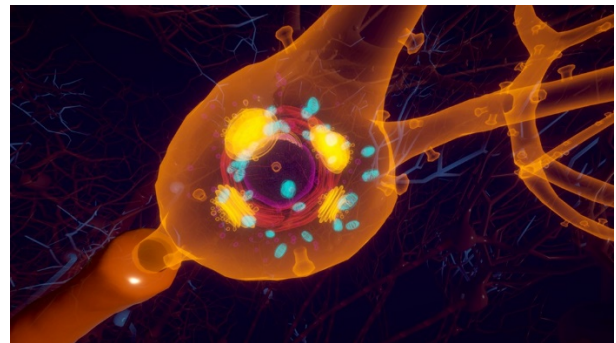


Figure 2: Neural Network rendering

- Hippocampus Neural Network – Clarity project

3 UNIQUE VISUALISATION

Sometimes it is hard to visualize what things look like in reality, so a bit of artistic approach was needed. Nobody has pictured the whole Observable Universe, neither anybody has seen the Electron and its color (as it has no color).

Visual representation of many elements shown in the project is unique and has never been displayed in a similar way before in an interactive environment. The renderings in the project are pushing limits of VR-ready graphics cards, from millions of particles to various rendering techniques used to show different cosmic phenomena.

4 EDUCATION AND INTERACTIVITY

The application contains an interactive element where the player can discover and interact with various objects and get detailed information about them. This feature adds a fun element to this

educational experience. So far the application has been received with very positive feedback, from pure amazement to learning something new about the world around us.

ACKNOWLEDGMENTS

The music used in the application was done by The Intangible. Various other sound elements were done by Andrej Cvicela.